

# Straight Leg Raises: The Foundation Exercise for Knee Osteoarthritis

Straight leg raises strengthen the quadriceps without bending the knee — the safest entry-point exercise for knee OA. Step-by-step guide with progression.

By **Living With Arthritis Editorial Team** · Reviewed by **Clinical Review Board** (HCPC-registered physiotherapists, BDA-registered dietitians) · 2026-05-30 · [livingwitharthritis.org.uk](https://livingwitharthritis.org.uk)

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## What this guide covers



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This piece summarises what the current evidence says about **straight leg raises**: the foundation exercise for knee osteoarthritis, the practical steps that help most, and the red flags that mean you should speak to your GP or physiotherapist. It is written for people living in the UK and reviewed by our clinical team.

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## The underlying mechanism

Arthritis pain rarely has a single cause. Joint cartilage, synovial fluid, ligaments, surrounding muscle and the nervous system all contribute. Understanding which of these is driving your specific symptom is the first step toward targeted relief. For an in-depth condition overview, see our [\[related guide\]\(/exercises/strength-training-for-knee\)](/exercises/strength-training-for-knee).

Inflammatory and degenerative forms behave differently. Inflammatory arthritis (rheumatoid, psoriatic, ankylosing spondylitis) typically worsens with rest and improves with gentle movement. Degenerative arthritis (osteoarthritis) typically worsens with prolonged use. Knowing which pattern matches your day-to-day experience helps you choose the right management approach.



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## What helps — by strength of evidence

1. **Movement and graded exercise.** NICE NG226 (2022) and Cochrane reviews place structured exercise as a first-line treatment. Aim for 150 minutes of moderate activity per week, split into 20–30 minute sessions. 2. **Anti-inflammatory eating.** A Mediterranean-style pattern — oily fish twice a week, plenty of vegetables, olive oil, whole grains — is linked to reduced pain scores in OA and RA trials. 3. **Weight management.** A 5 percent reduction in body weight reduces knee-OA symptoms measurably; a 10 percent reduction can match the effect of some medications. 4. **Heat and cold.** Heat eases stiffness; cold eases acute swelling. Twenty-minute applications, with a towel barrier. 5. **Sleep hygiene.** Pain perception rises sharply on fewer than six hours of sleep. Prioritise a consistent bedtime.

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## A practical seven-day starting plan

Begin with the smallest dose that feels manageable — five minutes of movement, one anti-inflammatory meal a day, a 10 pm screen-off rule. Track only three things weekly: pain on waking (0–10), longest pain-free interval, and one functional win (e.g., climbed stairs without grabbing the rail). Small, consistent inputs outperform big sporadic efforts.



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## When to seek clinical advice

Book an appointment with your GP or a first-contact physiotherapist if you notice: sudden hot, red, swollen joints; unexplained weight loss alongside joint pain; morning stiffness lasting longer than an hour for more than two weeks; or new numbness or weakness. These can signal inflammatory disease that benefits from early treatment.

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## Frequently asked questions

**\*\*Will this approach work alongside my medication?\*\*** Yes. Exercise, diet and pacing complement prescribed treatments and often improve their effectiveness. Always check with your prescriber before stopping any medication.

**\*\*How long until I notice a difference?\*\*** Most people report meaningful improvement at 6–12 weeks of consistent practice. Earlier wins (less morning stiffness, better sleep) often appear within 2–3 weeks.

**\*\*Is this safe if I have other conditions?\*\*** The principles above are broadly safe for adults with stable cardiovascular, metabolic and respiratory conditions. If you are unsure, ask your GP for a referral to a first-contact physiotherapist for a tailored plan.

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\*This article is general health information, not medical advice. Always consult a qualified healthcare professional for decisions about your individual care.\*

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## Q&A

**Question: What is the key thing to know about straight leg raises: the foundation exercise for knee osteoa?**

**Short answer:** The guide explains that knee osteoarthritis pain can involve cartilage, fluid, ligaments, muscle and the nervous system, and that management depends on symptom patterns. It recommends graded movement, anti-inflammatory eating, weight management, heat or cold, sleep hygiene, and small weekly tracking goals. It advises seeking GP or physiotherapy help for hot swollen joints, weight loss, prolonged stiffness, numbness or weakness.

**Question: When should I seek professional advice?**

**Short answer:** If symptoms persist or worsen, pain is severe and unresponsive to usual relief, or you develop new symptoms, contact your GP, physiotherapist or rheumatology team promptly.

**Question: Where can I get free arthritis support in the UK?**

**Short answer:** Arthritis UK: 0800 5200 520. NRAS: 0800 298 7650. Free tools and guides at [livingwitharthritis.org.uk](https://livingwitharthritis.org.uk).

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General information only — not medical advice. Consult your GP or specialist before changing medication or starting a new programme. Emergency: 999. NHS advice: 111. Living With Arthritis · [livingwitharthritis.org.uk](http://livingwitharthritis.org.uk)